Problem 4.6

At a certain rate of simple interest \$1,000 will accumulate to \$1,110 after a certain period of time. Find the accumulated value of \$500 at a rate of simple interest three fourths as great over twice as long a period of time.

Solution

For simple interest, the accumulation function is

$$a(t) = 1 + it.$$

From the first part of the problem,

$$1110 = 1000(1 + it).$$

Therefore,

$$\frac{1110}{1000} = 1 + it,$$

so

$$it = \frac{1110}{1000} - 1.$$

Now the new investment is \$500, the rate is $\frac{3}{4}i$, and the time is $2t$. Hence,

$$X = 500\left(1 + \frac{3}{4}i(2t)\right).$$

Simplifying,

$$X = 500\left(1 + \frac{3}{2}it\right).$$

Substituting the value of it,

$$X = 500\left(1 + \frac{3}{2}\left(\frac{1110}{1000} - 1\right)\right).$$

Thus,

$$X = 582.5.$$